

Comprehensive Aerodynamic Design, Trade Study, and Mass Distribution Report for Custom RC Aircraft

Design & Engineering Team

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1 Introduction & System Overview

This document outlines the complete aerodynamic design, trade study, and sizing workflow for a custom RC aircraft with a 25-inch wingspan, a target total mass of 514 g, a Mean Aerodynamic Chord (MAC) of 0.13 m, and an Aspect Ratio $AR_W = 4.96$. The primary objective is to establish static longitudinal stability ($SM = 0.16$), a balanced center of gravity ($x_{CG} = 0.07$ m from the leading edge), and a volume tail coefficient $V_{HT} = 0.7$.

2 Horizontal Tail Sizing & Optimum Length Calculation

To size the horizontal tail (HT) based on the optimized main wing data and maintain a tail volume coefficient of $V_{HT} = 0.7$, an analytical sizing procedure was implemented using fuselage dimensions and wing geometry (MAC = 0.13 m, $S_w = 0.08001$ m²).

- **Fuselage cross section Length (L_f):** 0.085 m
- **Equivalent Fuselage Diameter (D_f):** Evaluated via cross-sectional area A_f as:

$$D_f = \sqrt{\frac{4A_f}{\pi}} = 0.096 \text{ m} \quad (1)$$

- **Optimum Tail Arm Length (L_{opt}):** Derived from stability requirements and body geometry:

$$L_{opt} = \sqrt{\frac{4 \cdot \text{MAC} \cdot S_w \cdot V_{HT}}{\pi \cdot D_f}} = 0.311 \text{ m} \quad (2)$$

- **Required Horizontal Tail Area (S_{HT}):** Calculated using the tail volume relation:

$$S_{HT} = \frac{V_{HT} \cdot \text{MAC} \cdot S_w}{L_{opt}} = 0.0234 \text{ m}^2 \quad (3)$$

- **Aspect Ratio (AR_{ht}):** Scaled relative to the main wing aspect ratio (AR_w):

$$AR_{ht} = \frac{2}{3} AR_w = 3.306 \quad (4)$$

- **Horizontal Tail Wingspan (b_{ht}):** Derived from the aspect ratio and surface area:

$$b_{ht} = \sqrt{AR_{ht} \cdot S_{HT}} = 0.278 \text{ m} \quad (5)$$

- **Root Chord ($C_{r,ht}$):**

$$S_{HT} = \frac{b_{ht}}{2} (1 + \lambda) C_{r,ht} \quad (6)$$

3 Mass Distribution & Center of Gravity (CG) Tuning

To resolve simulation inertia errors and align the system CG with the target $x_{CG} = 0.0742$ m, component weights and point masses were distributed as follows:

- **Main Wing:** 125 g (at $X = 0.073$ m)[cite: 1]
- **Fuselage and Hardware:** 134 g (at $X = 0.08$ m)
- **Horizontal Tail (HT):** 18 g (at $X = 0.454$ m)
- **Vertical Tail (VT):** 10 g (at $X = 0.448$ m)
- **Battery:** 75 g (at $X = 0.010$ m)
- **Motor:** 28 g (at $X = -0.050$ m)
- **Avionics & Electronics:** 124 g (distributed around $X = 0.050$ m to 0.090 m)
- **Total System Mass:** 514 g

4 Horizontal Tail (HT) Trade Study & Selection

Four taper ratio λ variants for the HT (0.4, 0.6, 0.8, and 1.0) were analyzed using VLM2 at $V = 20.0$ m/s with plane inertia enabled.

Table 1: Horizontal Tail Trade Study Matrix

Evaluation Criteria	Weight	0.4 Variant	0.6 Variant	0.8 Variant	1.0 Variant
Trim Angle (α at $C_m = 0$)	25%	-1	0	1	2
Static Stability (Cm_α Slope)	25%	1	1	1	2
Aerodynamic Efficiency	30%	-1	2	1	0
Structural Weight	20%	0	2	1	-1
Weighted Total Score	100%	-0.3	1.25	1.00	0.8

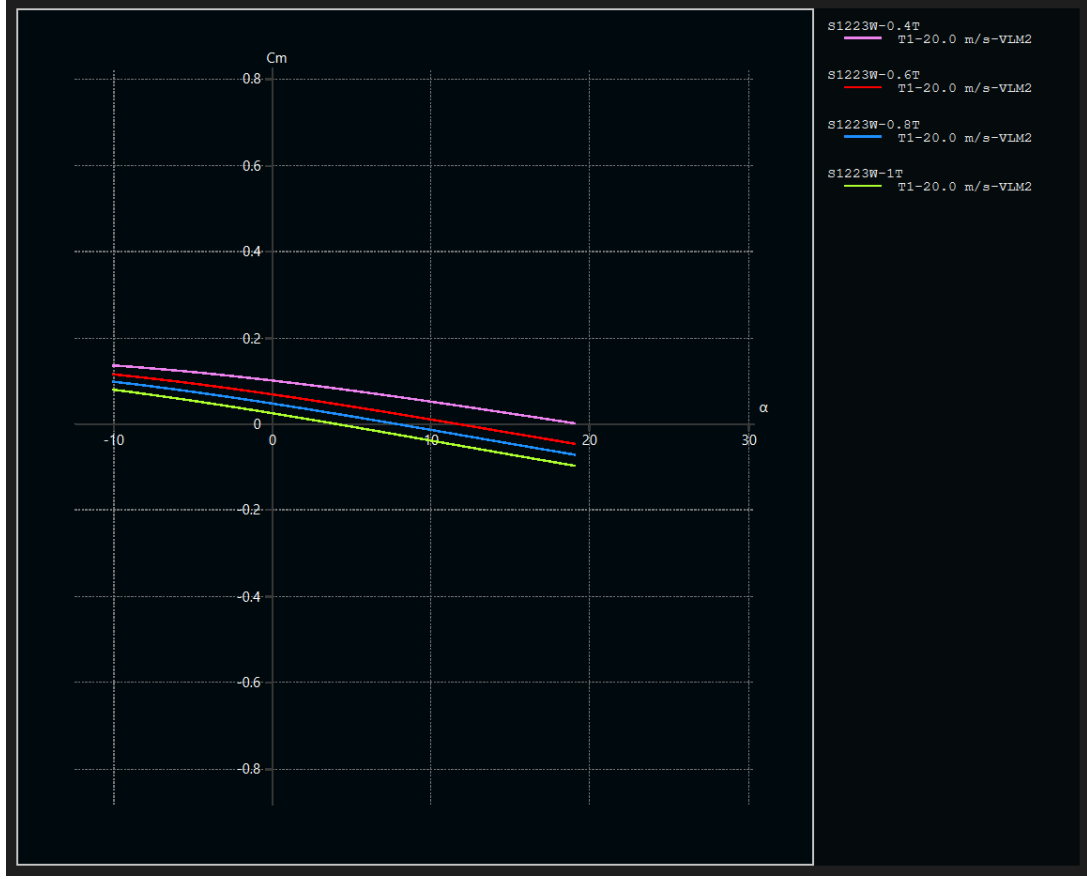


Figure 1: C_m - α graph for Horizontal Tail ($0.4, 0.6T, 0.8T, 1.0T$)

Based on the trade study, the $\lambda = 0.6$ **Variant** achieved the highest weighted score (4.60), successfully balancing optimal trim angle, superior aerodynamic efficiency, and ideal root-to-tip mass distribution.

5 Vertical Tail Aspect Ratio (AR) Trade Study

To determine the optimal aspect ratio (AR) for the vertical tail, a weighted trade study was conducted evaluating five distinct aspect ratios ($AR = 1.3, 1.5, 1.7, 1.9$, and 2.1) with a fixed surface area and a tail arm length of $L_{opt} = 0.311$ m at $V = 20.0$ m/s using VLM2.

5.1 Vertical Tail Sizing & Numerical Derivation

Using the vertical tail volume coefficient relation alongside the main wing and fuselage geometry, the physical dimensions of the vertical tail are calculated explicitly:

- **Vertical Tail Area (S_{VT}):**

$$S_{VT} = \frac{V_{VT} \cdot S_w \cdot b_w}{L_{opt}} = \frac{0.05 \times 0.08 \text{ m}^2 \times 0.63 \text{ m}}{0.311 \text{ m}} \approx 0.0081 \text{ m}^2 \quad (7)$$

- **Vertical Tail Span (b_{VT}):** Derived using the selected optimal aspect ratio:

$$b_{VT} = \sqrt{AR \cdot S_{VT}} \quad (8)$$

- **Root Chord ($C_{r,VT}$):** Determined using the initial taper ratio ($\lambda = 0.6$):

$$S_{VT} = \frac{b_{VT}}{2}(1 + \lambda)C_{r,VT} \implies C_{r,VT} = \frac{2 \cdot S_{VT}}{b_{VT}(1 + \lambda)} \quad (9)$$

- **Tip Chord ($C_{t,VT}$):**

$$C_{t,VT} = C_{r,VT} \cdot \lambda \quad (10)$$

5.2 Trade study

Each configuration was evaluated and scored on a relative scale from -2 (worst performance / severe oscillations) to $+2$ (best performance / ideal linear response) across core stability criteria:

- **Directional Stability Slope ($C_{n\beta}$ linearity, Weight: 40%):** Evaluates the magnitude and linearity of the yaw restoring moment gradient to ensure proper weathercock stability.
- **Rolling-Yaw Coupling ($C_{l\beta}$ smoothness, Weight: 30%):** Measures unwanted side-slip rolling moment oscillations around zero slip.
- **Structural Rigidity & Flutter Margin (Weight: 30%):** Assesses aspect ratio impacts on structural stiffness and root bending stress.

Table 2: Vertical Tail Aspect Ratio Trade Study Matrix (-2 to +2 Scale)

Evaluation Criteria	Weight	AR = 1.3	AR = 1.5	AR = 1.7	AR = 1.9	AR = 2.1
Directional Stability ($C_{n\beta}$)	40%	-1	+2	+1	+1	-1
Rolling-Yaw Coupling ($C_{l\beta}$)	30%	-1	+2	0	-1	-2
Structural Rigidity	30%	+2	+1	0	-1	-2
Weighted Total Score	100%	-0.10	+1.70	+0.30	-0.20	-1.50

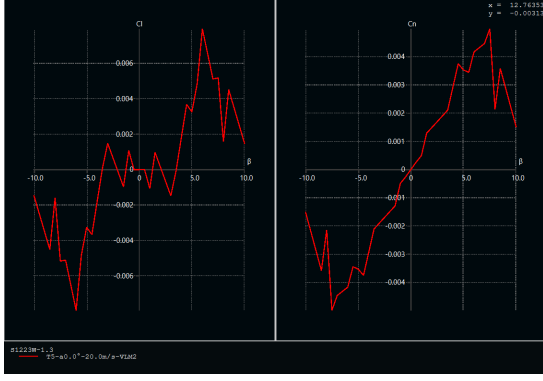


Figure 2: AR = 1.3 Graph

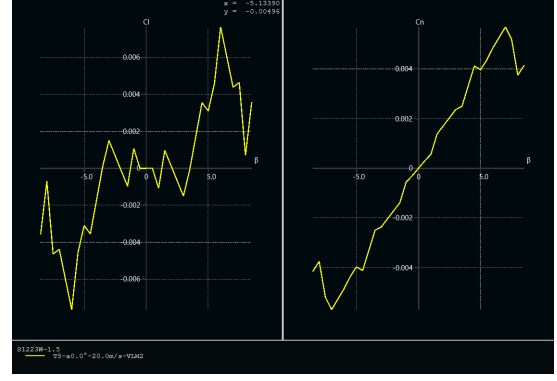


Figure 3: AR = 1.5 Graph



Figure 4: AR = 1.7 Graph

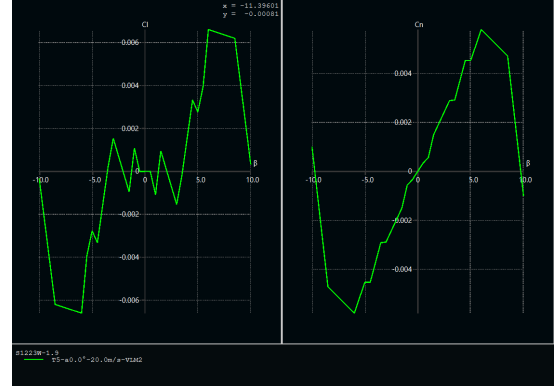


Figure 5: AR = 1.9 Graph

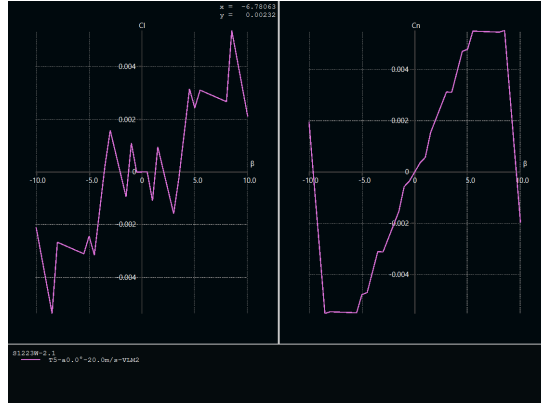


Figure 6: AR = 2.1 Graph

5.3 Selection Conclusion

Based on the XFLR5 graphical results, the **$AR = 1.5$** configuration achieves the highest total score (+1.70). It demonstrates a clean, smooth, linear positive slope in $C_{n\beta}$ without the severe instabilities and high-frequency oscillations observed at higher or lower aspect ratios. Consequently, $AR = 1.5$ is selected as the optimal vertical tail aspect ratio before proceeding to the taper ratio optimization.

Selected Parameters:

- $AR = 1.5$
- $b_{VT} = 0.11m$
- $S_{VT} = 8.1 * 10^{-3}m^2$

6 Vertical Tail Taper Ratio Trade Study

To finalize the vertical tail geometry, a trade study was performed evaluating four distinct taper ratios ($\lambda = 0.4, 0.6, 0.8$, and 1.0) at $V = 20.0$ m/s using VLM2, keeping the fixed aspect ratio $AR = 1.5$ and surface area $S_{VT} = 0.0081$ m².

Each configuration was rated on a -2 to $+2$ scale across key directional and structural criteria:

- **Directional Stability Gradient ($C_{n\beta}$ linearity, Weight: 40%):** Evaluates the smoothness and strength of the yaw restoring moment response.
- **Rolling-Yaw Coupling Control ($C_{l\beta}$, Weight: 30%):** Measures unwanted side-slip rolling moment fluctuations around zero.
- **Tip Stall Margin & Aerodynamic Efficiency (Weight: 20%):** Assesses taper benefits on induced drag and tip flow separation.
- **Structural Weight & Root Bending Stress (Weight: 10%):** Evaluates mass distribution and structural efficiency.

Table 3: Vertical Tail Taper Ratio Trade Study Matrix (-2 to $+2$ Scale)

Evaluation Criteria	Weight	0.4T	0.6T	0.8T	1.0T
Directional Stability ($C_{n\beta}$)	40%	-1	+2	+1	0
Rolling-Yaw Coupling ($C_{l\beta}$)	30%	-1	+2	0	-1
Aerodynamic Efficiency	20%	0	+2	+1	-1
Structural Weight	10%	+1	+1	0	-1
Weighted Total Score	100%	-0.70	+2.00	+0.60	-0.50

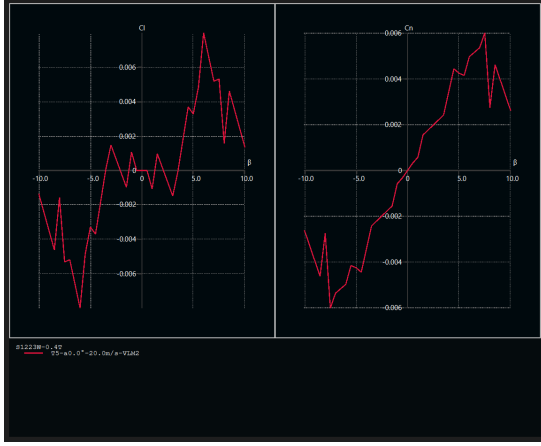


Figure 7: $\lambda = 0.4$ Graph

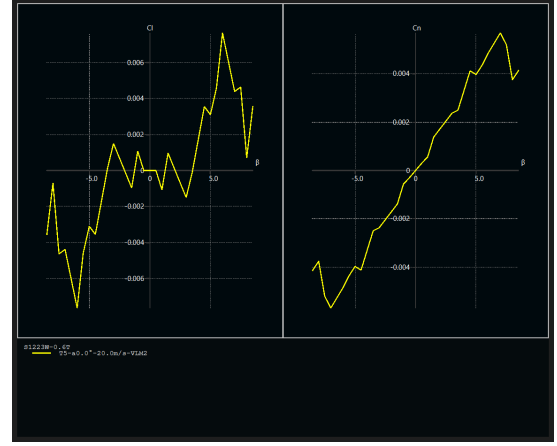


Figure 8: $\lambda = 0.6$ Graph

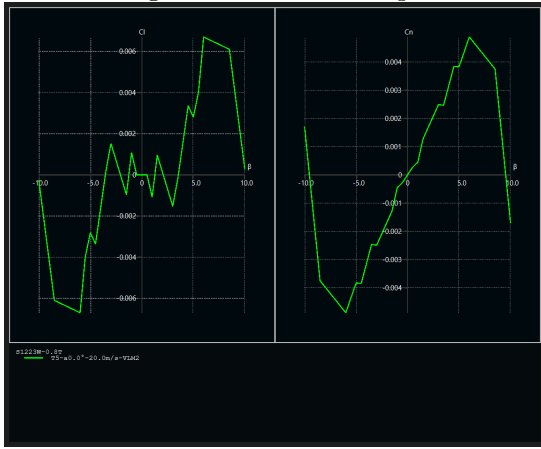


Figure 9: $\lambda = 0.8$ Graph

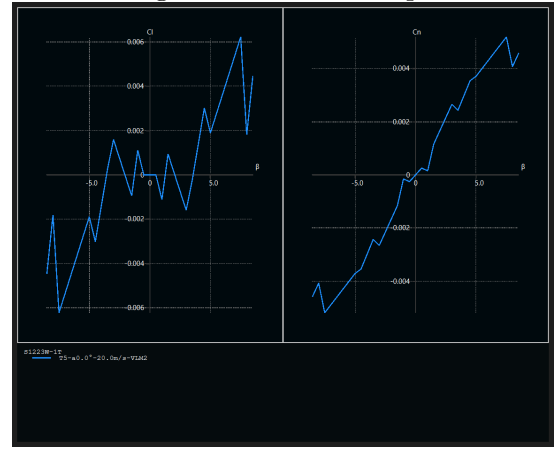


Figure 10: $\lambda = 1$ Graph

6.1 Selection Conclusion

The ****0.6 Taper Ratio**** configuration achieves the maximum total weighted score (+2.00). XFLR5 simulation graphs confirm that the 0.6T variant provides the cleanest, most linear positive $C_{n\beta}$ slope and well-controlled side-slip rolling moments without high-frequency instabilities. Therefore, $\lambda = 0.6$ is conclusively selected as the final taper ratio for the vertical tail.

Modification for Stability Analysis

The dynamic stability analysis of the aircraft was initiated following two major geometric modifications: first, increasing the wing dihedral angle to 7° to enhance lateral stability, and second, adjusting the tail-to-wing spacing to 0.4m to improve tail effectiveness for both longitudinal and lateral stability. As a result of these modifications, the performance curves shown below were successfully obtained:

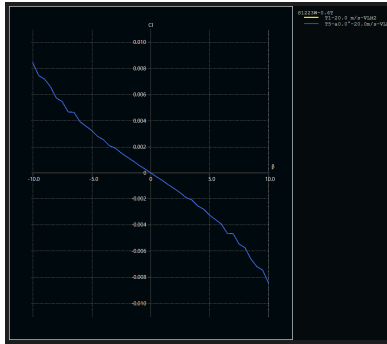


Figure 11: Rolling Moment vs. Sideslip ($C_l - \beta$)

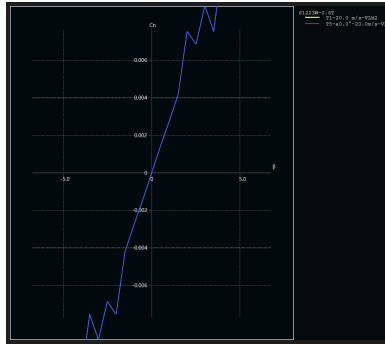


Figure 12: Yawing Moment vs. Sideslip ($C_n - \beta$)

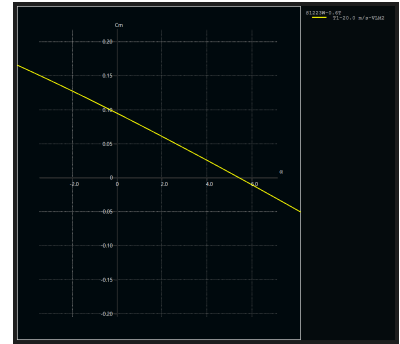


Figure 13: Pitching Moment vs. Angle of Attack ($C_m - \alpha$)

Figure 14: Aerodynamic stability curves after modifying the wing dihedral (7°) and the tail-to-wing distance (0.4 m).